Thm. V is a vector space over F. $\dim(V) = n$.
$\mathcal{B} = \{\bar{\alpha}_1, \dots, \bar{\alpha}_n\}$, $\mathcal{B}' = \{\bar{\alpha}_1', \dots, \bar{\alpha}_n'\}$ ordered bases for V.

$T \in L(V)$. $P = [P_1 \dots P_n] \in F^{m \times n}$ with
$P_j = [\bar{\alpha}_j']_{\mathcal{B}}$. Then, $[T]_{\mathcal{B}'} = P^{-1} [T]_{\mathcal{B}} P$.

Alternatively, if $U \in L(V)$ & U^{-1} exists
s.t. $U\bar{\alpha}_j = \bar{\alpha}_j' \ \forall j \in \{1, \dots, n\}$, then

$$[T]_{\mathcal{B}'} = [U]_{\mathcal{B}}^{-1} [T]_{\mathcal{B}} [U]_{\mathcal{B}}.$$

$A_{n\times n}$ & $B_{n \times n}$ are square matrices over F.
B is similar to A if $\exists\, P, P^{-1} \in F^{n \times n}$ st.

$$B = P^{-1} A P.$$

Example: $T \in L(\mathbb{R}^2)$, $T(x_1, x_2) = (x_1, 0)$.

$\mathcal{B} = \{\bar{\epsilon}_1, \bar{\epsilon}_2\}$. $[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$\mathcal{B}' = \{\bar{\epsilon}_1' = (1,1), \bar{\epsilon}_2' = (2,1)\}$. $\bar{\epsilon}_1' = \bar{\epsilon}_1 + \bar{\epsilon}_2$
$\bar{\epsilon}_2' = 2\bar{\epsilon}_1 + \bar{\epsilon}_2$

Then, $P = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$, $P^{-1} = \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$

$$[T]_{\mathcal{B}'} = P^{-1} [T]_{\mathcal{B}} P = \begin{bmatrix} -1 & -2 \\ 1 & 2 \end{bmatrix}.$$

$T\bar{\epsilon}_1' = (1,0) = -\bar{\epsilon}_1' + \bar{\epsilon}_2'$

$T\bar{\epsilon}_2' = (2,0) = -2\bar{\epsilon}_1' + 2\bar{\epsilon}_2'$.

Representation of transformations by Matrices.

- $T: V \to W$ a linear transformation.

Ordered bases $\mathcal{B}_V = \{\bar{\alpha}_1, \dots, \bar{\alpha}_n\}$

$\dim(V) = n$ $\mathcal{B}_W = \{\beta_1, \dots, \beta_m\}$

$\dim(W) = m$

Each of the n vectors $T\bar{\alpha}_j$ is uniquely expressable as $T\bar{\alpha}_j = \sum_{i=1}^{m} A_{ij} \bar{\beta}_i$ ——— (1)

$[T\bar{\alpha}_j]_{\mathcal{B}_W} = (A_{1j}, \dots, A_{mj}) \to$ jth column of A.

T is determined by $\underbrace{A_{m \times n}}_{\text{matrix of } T \text{ relative to } \mathcal{B}_V \,\&\, \mathcal{B}_W} = [A_{ij}]_{ij}$

If $\bar{\alpha} = \sum_{j=1}^{n} x_j \bar{\alpha}_j$,

$$T\bar{\alpha} = T\left(\sum_{j=1}^{n} x_j \bar{\alpha}_j\right) = \sum_{j=1}^{n} x_j (T\bar{\alpha}_j)$$

$$= \sum_{j=1}^{n} x_j \left(\sum_{i=1}^{m} A_{ij} \bar{\beta}_i\right) = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} A_{ij} x_j\right) \bar{\beta}_i$$

$$[\bar{\alpha}]_{\mathcal{B}_V} = (x_1, \dots, x_n)$$

$$[T\bar{\alpha}]_{\mathcal{B}_W} = A[\bar{\alpha}]_{\mathcal{B}_V}.$$

- If $A_{m \times n}$ is a matrix over F, then

$$T\left(\sum_{j=1}^{n} x_j \bar{\alpha}_j\right) = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} A_{ij} x_j\right) \bar{\beta}_i$$

defines $T: V \to W$, a linear transf. A (relative to $\mathcal{B}_V, \mathcal{B}_W$).

Thm: V & W be vector spaces over F. $\dim(V)=n$, $\dim(W)=m$. $\mathcal{B}_V$ & $\mathcal{B}_W$ are ordered bases of V & W, respectively. For each linear transf. $T: V \to W$, $\exists$ an $m \times n$ matrix A over F s.t.

$$[T\bar{\alpha}]_{\mathcal{B}_W} = A[\bar{\alpha}]_{\mathcal{B}_V}$$

$\forall \bar{\alpha} \in V$. Furthermore, $T \to A$ is a 1:1 correspondence betn the set of all linear transformations from $V \to W$ & the set of all $m \times n$ matrices over F.

$A_{m \times n} \to$ matrix of T relative to $\mathcal{B}_V, \mathcal{B}_W$
($T: V \to W$, $\dim V = n$, $\dim W = m$).

$$T\bar{\alpha}_j = \sum_{i=1}^{n} A_{ij} \bar{\beta}_i \quad \forall \bar{\alpha}_j \in \mathcal{B}_V.$$

$$A = [A_1 \ \dots \ A_n]$$

$$A_j = [T\bar{\alpha}_j]_{\mathcal{B}_W}$$

If $U: V \to W$ is another linear transf. $B = [B_1 \ \dots \ B_n]$ is matrix of U relative to $\mathcal{B}_V, \mathcal{B}_W$. then $cA + B$ is matrix of $cT + U$ relative to $\mathcal{B}_V, \mathcal{B}_W$.

$$cA_j + B_j = c[T\bar{\alpha}_j]_{\mathcal{B}_W} + [U\bar{\alpha}_j]_{\mathcal{B}_W} = [cT\bar{\alpha}_j + U\bar{\alpha}_j]_{\mathcal{B}_W}$$
$$= [(cT+U)\bar{\alpha}_j]_{\mathcal{B}_W}$$

Thm. V & W are vector spaces over F.
$\dim V = n$, $\dim W = m$.
$\mathcal{B}_V$: ordered basis in V
$\mathcal{B}_W$: ordered basis in W
For each pair $\mathcal{B}_V, \mathcal{B}_W$, the f^n which assigns to a linear map $T: V \to W$ its matrix relative to $\mathcal{B}_V, \mathcal{B}_W$ is an isomorphism b/w the space $L(V,W)$ and the space $F^{m\times n}$ (set of all $m\times n$ matrices over F).

For $T \in L(V)$, $A_{n\times n}$ is matrix of T w.r.t. ordered basis $\mathcal{B}_V$. (we take $W=V$, $\mathcal{B}_W = \mathcal{B}_V$).

$$T\bar{\alpha}_j = \sum_{i=1}^{n} A_{ij}\bar{\alpha}_i \quad , j \in \{1, \dots, n\}.$$

Note: Matrix A representing T depends on $\mathcal{B}_V$.
There is a representing matrix of T in each ordered basis for V.

$A = [T]_{\mathcal{B}_V} \to$ matrix A represents T w.r.t. $\mathcal{B}_V$.

$$[T\bar{\alpha}]_{\mathcal{B}_V} = [T]_{\mathcal{B}_V}[\bar{\alpha}]_{\mathcal{B}_V}.$$

Examples ① $V = F^{n\times 1}$ $W = F^{m\times 1}$ $A_{m\times n} \in F^{m\times n}$.

$T : V \to W$ $(T \in L(V,W))$

$T(X) = AX$.

$\mathcal{B}_V$: std basis in $F^{n\times 1}$ $\{X_1, \ldots, X_n\}$

$\mathcal{B}_W$: ~~std~~ std basis in $F^{m\times 1}$ $\{Y_1, \ldots, Y_m\}$

$[T]_{\mathcal{B}_V, \mathcal{B}_W} = A$. $AX_j \to j^{th}$ column of A.

$X_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \to i^{th}$ row $\quad Y_j = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \to j^{th}$ row.

⑩ $T : F^2 \to F^2$, $T(x_1, x_2) = (x_1, 0)$

$\mathcal{B} = \{\underset{(1,0)}{\epsilon_1}, \underset{(0,1)}{\epsilon_2}\}$ $\quad T\epsilon_1 = (1,0) = 1\,\epsilon_1 + 0\,\epsilon_2$

$T\epsilon_2 = (0,0) = 0\,\epsilon_1 + 0\,\epsilon_2$

$[T]_{\mathcal{B}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

V, W, Z vector spaces over F.
$\dim V = n$, $\dim W = m$, $\dim Z = p$.
$T: V \to W$ $(T \in L(V,W))$,
$U: W \to Z$ $(U \in L(W,Z))$.
$\mathcal{B}_V = \{\bar{\alpha}_1, \ldots, \bar{\alpha}_n\}$, $\mathcal{B}_W = \{\bar{\beta}_1, \ldots, \bar{\beta}_m\}$ | ordered bases
$\mathcal{B}_Z = \{\bar{\gamma}_1, \ldots, \bar{\gamma}_p\}$

$$A = [T]_{\mathcal{B}_V, \mathcal{B}_W}, \quad B = [U]_{\mathcal{B}_W, \mathcal{B}_Z}.$$

Then, $C = [UT]_{\mathcal{B}_V, \mathcal{B}_Z} = BA$;

for, if $\bar{\alpha} \in V$,

$$[T\bar{\alpha}]_{\mathcal{B}_W} = A[\bar{\alpha}]_{\mathcal{B}_V},$$

$$[U(T\bar{\alpha})]_{\mathcal{B}_Z} = B[T\bar{\alpha}]_{\mathcal{B}_W} = BA[\bar{\alpha}]_{\mathcal{B}_V}.$$

Hence, by definition & uniqueness of the representing matrix, we must have

$$C = BA.$$

$$(UT)\bar{\alpha}_j = U(T\bar{\alpha}_j) = U\left(\sum_{k=1}^{m} A_{kj}\bar{\beta}_k\right) = \sum_{k=1}^{m} A_{kj}(U\bar{\beta}_k)$$

$$= \sum_{k=1}^{m} A_{kj} \sum_{i=1}^{p} B_{ik}\bar{\gamma}_i = \sum_{i=1}^{p}\left(\sum_{k=1}^{m} B_{ik}A_{kj}\right)\bar{\gamma}_i$$

$$\Rightarrow \quad C_{ij} = \sum_{k=1}^{m} B_{ik}A_{kj}.$$

Thm. V, W, Z are finite-dim vector spaces over F.
$T \in L(V,W)$, $U \in L(W,Z)$.
$\mathcal{B}_V, \mathcal{B}_W, \mathcal{B}_Z$: ordered bases of V, W, Z, resp.
If $A = [T]_{\mathcal{B}_V, \mathcal{B}_W}$, $B = [U]_{\mathcal{B}_W, \mathcal{B}_Z}$,
then $[U \circ T]_{\mathcal{B}_V, \mathcal{B}_Z} = BA$.

Note: Above thm provides proof that matrix multiplication is associative.

When $T \in L(V)$, $U \in L(V)$,

$$[UT]_{\mathcal{B}_V} = [U]_{\mathcal{B}_V} [T]_{\mathcal{B}_V}.$$

Correspondence which $\mathcal{B}_V$ determines b/w operators & matrices is not only a vector space isomorphism but also preserves products. As a consequence, $T \in L(V)$ is invertible iff $[T]_{\mathcal{B}_V}$ is invertible.

~~For, $\mathbb{1}$ is represented~~

Identity operator $\mathbb{1}$ is represented by $\mathbb{1}$ in any ordered basis, & thus

$$UT = TU = \mathbb{1}$$

is equivalent to

$$[U]_{\mathcal{B}_V} [T]_{\mathcal{B}_V} = [T]_{\mathcal{B}_V} [U]_{\mathcal{B}_V} = \mathbb{1}.$$

When T is invertible, $[T^{-1}]_{\mathcal{B}_V} = [T]^{-1}_{\mathcal{B}_V}$.

What happens to A when ordered bases are changed?

Consider $T \in L(V)$. $\dim(V) = n$

$\mathcal{B} = \{\bar{\alpha}_1, \ldots, \bar{\alpha}_n\}$, $\mathcal{B}' = \{\bar{\alpha}_1', \ldots, \bar{\alpha}_n'\}$

$\mathcal{B}, \mathcal{B}' \rightarrow$ ordered bases for V.

There is a unique (invertible) $n \times n$ matrix P s.t. $[\bar{\alpha}]_{\mathcal{B}} = P[\bar{\alpha}]_{\mathcal{B}'}$, $\forall \bar{\alpha} \in V$. — (I)

$P_{n\times n} = [P_1 \; P_2 \; \ldots \; P_n]$, $P_j = [\bar{\alpha}_j']_{\mathcal{B}}$.

By definition, $[T\bar{\alpha}]_{\mathcal{B}} = [T]_{\mathcal{B}} [\bar{\alpha}]_{\mathcal{B}}$, — (II)

then $[T\bar{\alpha}]_{\mathcal{B}} = P[T\bar{\alpha}]_{\mathcal{B}'}$ — (III) (using (I) & (II)).

From (I), (II), (III), $[T]_{\mathcal{B}} P[\bar{\alpha}]_{\mathcal{B}'} = P[T\bar{\alpha}]_{\mathcal{B}'}$

or, $P^{-1}[T]_{\mathcal{B}} P[\bar{\alpha}]_{\mathcal{B}'} = [T\bar{\alpha}]_{\mathcal{B}'}$

$$\Rightarrow [T]_{\mathcal{B}'} = P^{-1}[T]_{\mathcal{B}} P.$$

$\exists$ a unique linear operator U that maps $\mathcal{B}$ to $\mathcal{B}'$, $U\bar{\alpha}_j = \bar{\alpha}_j'$, $\forall j \in \{1, \ldots, n\}$.

U is invertible (maps basis to basis).

$P = [U]_{\mathcal{B}}$ by def^n | For, P is defined by $\bar{\alpha}_j' = \sum_{i=1}^{n} P_{ij}\bar{\alpha}_i$

$\because U\bar{\alpha}_j = \bar{\alpha}_j'$, $U\bar{\alpha}_j = \sum_{i=1}^{n} P_{ij}\bar{\alpha}_i$.

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